Sapphire Capability and Transparency Manual

What is tested, what is optional, and what remains roadmap work.

This manual describes the currently tested implementation in this repository and intentionally avoids claiming production-scale or distributed infrastructure.

It is a local runtime reference for learning, experimentation, scripting, and bounded automation work, not a verified multi-node training or cloud deployment platform.

Tested areas

The repository tests cover language execution, file and process helpers, concurrency, agent planning, failure handling, persistence, memory retrieval, sandbox boundaries, CPU and ML runtimes, and optional CUDA reporting.

Optional dependencies

NumPy, PyTorch, psutil, requests, audio packages, AI services, NVIDIA drivers, and CUDA hardware are optional or environment-specific. These are not assumptions made for every installation.

Distributed and multi-node claims

Distributed feature names describe planning and generation artifacts. They do not prove that a cluster launched, trained, recovered, or delivered measured throughput in a real multi-node environment.

Roadmap

Native compilation, stronger isolation guarantees, event-driven agent declarations, DAG scheduling, durable neural memory, verified multi-node training, and expanded security validation remain future work.

Current status summary

Sapphire version 1.0.5 is a Python-based local runtime and language with a working interpreter, bundled tools, and optional integrations. It is appropriate for learning, local automation, and experimental AI workflows. It is not presented as a distributed training stack, a production security-certified platform, or a globally managed service.

Results vary by machine, Python version, OS configuration, library availability, and workload. The documentation here reflects what is actually exercised in this repository and what remains a roadmap item.

Version 1.0.5. Reproduce local measurements with benchmarks/benchmark_runtime.py. Values depend on the machine and environment.